The properties of diamagnetic materials are

i) Magnetic susceptibility is negative.

ii) Relative permeability is slightly less than unity.

iii) The magnetic field lines are repelled or expelled by diamagnetic materials when placed in a magnetic field.

iv) Susceptibility is nearly temperature independent.

Note

Superconductors are perfect diamagnetic materials. The expulsion of magnetic flux from a superconductor during its transition to the superconducting state is known as Meissner effect. (see figure 3.18)

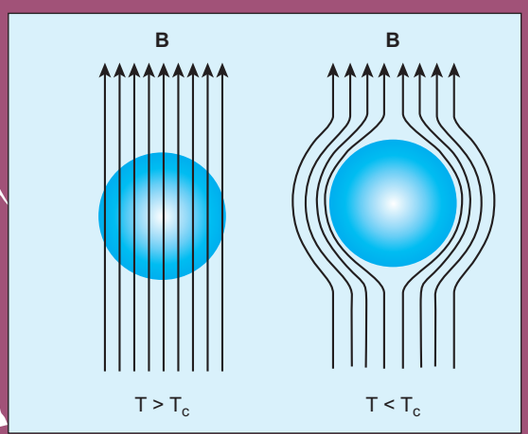


Figure 3.18 Meissner effect – superconductors behave like perfect diamagnetic materials below transition temperature T_C.

(b) Paramagnetic materials

In some magnetic materials, each atom or molecule has net magnetic dipole moment which is the vector sum of orbital and spin magnetic moments of electrons. Due to the random orientation of these magnetic moments, the net magnetic moment of the materials is zero.

DO YOU KNOW?

Magnetic levitated train

Magnetic levitated train is also called Maglev train. This train floats few centimetres above the guideway because of electromagnet used. Maglev train does not need wheels and also achieve greater speed. The basic mechanism of working of Maglev train involves two sets of magnets. One set is used to repel which makes train to float above the track and another set is used to move the floating train ahead at very great speed. These trains are quieter, smoother and environmental friendly compared conventional trains and have potential for moving with much higher speeds with technology in future.



In the presence of an external magnetic field, the torque acting on the atomic dipoles will align them in the field direction. As a result, there is net magnetic dipole moment induced in the direction of the applied field. The induced dipole moment is present as long as the external field exists.

When placed in a non-uniform magnetic field, the paramagnetic materials will have a tendency to move from weaker to stronger part of the field. Materials which exhibit weak magnetism in the direction of the applied field are known as paramagnetic materials.

12th Physics_Vol-1_Unit 3_English Medium.indd 144 23-12-2022 15:02:05